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FBCSP and Adaptive Boosting for Multiclass Motor Imagery BCI Data Classification: A Machine Learning Approach

Rig Das
Department of Health Technology
Technical University of Denmark
Ørsteds Plads, 2800 Kgs. Lyngby, Denmark.
rigdas@dtu.dk

Paula S. Lopez
Department of Health Technology
Technical University of Denmark
Ørsteds Plads, 2800 Kgs. Lyngby, Denmark.

Muhammad Ahmed Khan
Department of Health Technology
Technical University of Denmark
Ørsteds Plads, 2800 Kgs. Lyngby, Denmark.
mahkh@dtu.dk

Helle K. Iversen
Department of Neurology
University of Copenhagen
Rigshospitalet, 2600, Glostrup, Denmark
helle.klingenberg.iversen@regionh.dk

Sadasivan Puthusserypady
Department of Health Technology
Technical University of Denmark
Ørsteds Plads, 2800 Kgs. Lyngby, Denmark.
sapu@dtu.dk

***Abstract*—Classification of non-stationary electroencephalogram (EEG) data are of utmost importance for brain-computer interface (BCI) technology. This paper proposes a robust multiclass motor imagery (MI) BCI data classification technique. It is based on filter bank common spatial patterns (FBCSP) and AdaBoost classification technique. The method is tested on the 4-class MI BCI competition IV dataset 2a and the results show superior performance compared to the current state-of-the-art performances. This paper also analyzes different frequency sub-bands for the MI EEG data, in order to find the best sub-band which contains the most significant features for distinguishing different MI tasks.**

***Index Terms*—Brain computer interface (BCI), filter-bank common spatial patterns (FBCSP), motor imagery (MI), Adaptive boosting (AdaBoost)**

I. Introduction

Over the past two decades, brain computer interface (BCI) have emerged as a medium for humans to interact with computers or other electronic equipment, without any practical use of sensory nerves and muscles and by using only the electroencephalographic (EEG) signals [1]. The attributes of the acquired EEG data depends upon the acquisition protocol that is employed for obtaining a particular subject's behavior. Various task-specific responses are typically conveyed in terms of a small time-locked shifts in the electrical pulses of the brain which can be collected using different stimulation standards entailing sensory or cognitive audio/video stimuli [2].

This research work investigates the EEG signals that are generated from different motor imagery (MI) related tasks in order to automatically separate them [1], [3]. MI activates specific regions of the brain which is involved in the preparation and execution of a movements [4]. This produces automatic time-locked signals from the subjects' brain, which can be recorded using EEGs. Researchers have shown that MI is a suitable means for designing BCIs which can help disabled people to communicate as well as control assistive devices. However, it is worth mentioning over here that for generating MI-elicited EEG data, the prerequisites are both high attention and keenness of the subjects' to imagine specific movements. Moreover, these EEG signals can get corrupted by eye-blinking/movements and other types of non-cerebral signals. Therefore, extraction of particular features from EEG signals for the separation of various MI tasks become extremely difficult through manual or simple approaches.

In this paper, we use the BCI competition IV 2a dataset [5] in order to separate 4 different MI tasks: left and right hands, feet and tongue movements. FBCSP filters have been employed to extract features and then these features are used in adaptive boosting (AdaBoost) machine learning technique for classification. A set of pre-processing techniques for the EEG signals is also proposed to improve their signal to noise ratio. Moreover, this paper experimentally and systematically determines the particular frequency sub-band of the EEG signals that contributes significantly for the MI task separation.

II. MI-EEG Data Classification: State-of-the-Art Literature

A short summary of the state-of-the-art research works in the field of MI-task elicited EEG signal separation is presented in Table I. In [6], Ang et al. proposed a FBCSP based feature extraction technique for the first time on BCI competition IV 2a and 2b datasets. A Chebyshev Type II band-pass filter is used to decompose the EEG signals into 9 different frequency sub-bands: $[4-8]$, $[8-12]$, ..., $[36-40]$Hz. Classification of different MI tasks are performed using Naïve Bayesian Parzen

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TABLE I
OVERVIEW OF THE STATE-OF-THE-ART TECHNIQUES FOR MI TASK SEPARATION.

Paper	Year	Ch.s	Database	Sub-band (Hz)	Features	Classifier	Approach
Ang et al. [6]	2008	22	BCI IV 2a and 2b	$4-40$	FBCSP	NBPW	OVR,PW,DC
Gouy-Pialler et al. [7]	2010	22	BCI IV 2a	$5-35$	CSP by JAD	Regularized multinomial regression	OVR
Wang [8]	2011	22	BCI IV 2a	$5-35$	WPC and WPC/TI	Euclidean distance	OVR
Barachant et al. [9]	2012	22	BCI IV 2a	$8-30$	spatial covariance matrices	Riemannian geometry	OVR
Asensio-Cubero et al. [10]	2013	22	BCI IV 2a	$8-30$	LDB+CSP	LDA	OVR
Asensio-Cubero et al. [11]	2013	15	BCI IV 2a, Essex dataset	$8-30$	MRA	LDA	OVR
Kam et al. [12]	2013	22	BCI IV 2a	$4-40$	CSP	LDA and SVM	OVR
Yang et al. [13]	2015	22	BCI IV 2a	$4-40$	ACSP	CNN	OVR
Nicolas-Alonso et al. [1]	2015	22	BCI IV 2a	$4-40$	FBCSP	SRKDA and SUSS-SRKDA	OVR
He et al. [14]	2016	15	BCI IV 2a	$0.1-40$	common Bayesian network	SVM	OVR
Miao et al. [15]	2017	22	BCI IV 2a, III 4a	$4-40$	discriminative spatial-frequency-temporal patterns	WNBC	OVR
Gaur et al. [16]	2018	22	BCI IV 2a	$8-30$	SS-MEMD	Riemannian geometry	OVR

Window (NBPW) classifier. Being a multi-class problem, three different approaches have been proposed for the classification: (i) Divide-and-Conquer (DC), (ii) Pair-wise (PW)/one vs. one (OVO), and (iii) One vs. Rest (OVR). For the PW and OVR approach, kappa (κ) value of 0.572 and 0.569 are achieved respectively. This work has also won the BCI competition IV. Later, Gouy-Pialler et al. [7] and Wang [8] have used a frequency sub-band of [$5-35$]Hz for the MI-task separation and compared their performance for OVR approach. An average κ value of 0.50 is achieved in [7], where the authors have used joint approximate diagonalization (JAD) for spatial filtering with the help of maximum likelihood framework. Common spatial pattern (CSP) filtering is used along with JAD for feature extraction and regularized multinomial regression for classification purposes. Similarly in [8], weighted pairwise criterion (WPC) and WPC with temporal information (WPC/TI) are used for feature extraction from the MI-task elicited EEG signals and Euclidean distance as a classifier to separate different classes, resulting in a κ value of 0.58.

Barachant et al. [9] proposed a feature extraction method using spatial covariance matrix from the [$8-30$]Hz spectral filtered EEG data. Reimannian geometry is used for classification, which gave a κ value of 0.57. In 2013, Asensio-Cubero et al. published two articles on MI task classification [10], [11], where [$8-30$]Hz frequency sub-band is used for further processing. In [10], local discriminant bases (LDB) along with CSP is used to extract features and linear discriminant analysis (LDA) for classification. In comparison, [11] uses only 15 selected channels out of the 22 channels, which are mainly positioned in the fronto-central (FC), central (C) and central-parietal (CP) regions of the brain. A study using multi-resolution analysis (MRA) of EEG signals' feature extraction is performed while LDA is used for classification. An average κ value of 0.59 is achieved for this method.

Kam et al. [12], Yang et al. [13] and Nicolas-Alonso et al. [1] have used CSP, augmented CSP (ACSP) and FBCSP, respectively, for feature extraction. For classification, LDA and support vector machine (SVM) are used in [12], convolutional neural network (CNN) in [13] and spectral regression kernel discriminant analysis (SKRDA) and sequential updating semi-supervised SRKDA (SUSS-SRKDA) methods in [1]. Authors of [1] have been able to outperform all other previous performances by achieving an average κ value of 0.70.

In recent years, He et al. [14] have used 15 channels, similar to [11], to extract the MI features using common Bayesian network. Using SVM classifier, an average κ value of 0.66 was achieved. Miao et al. in [15], used both BCI IV 2a and III 4a datasets and [$4-40$]Hz sub-band for feature extraction. Discernibility of feature sets (DFS) benchmark for the optimization of spatial filters is proposed here and then by utilizing sparse regression, i.e., sparse time-frequency segment common spatial pattern (STFSCSP), CSP features on time-frequency domain are automatically extracted. Finally, a weighted Naïve Bayesian classifier (WNBC) is used for classification. In [16], Gaur et al. have exploited [$8-30$]Hz frequency sub-band to extract the MI features using subject specific multivariate empirical mode decomposition (SS-MEMD) method and then by applying Reimannian geometry for classification, an average κ value of 0.60 is achieved.

One of the recent articles by Razi et al. [3] showed that using Dempster-Shafer theory (DST), they have been able to achieve a κ value of 0.75 for OVR method. But to the best of our knowledge, we think that combining three different approaches, i.e. OVO, OVR and TVT (Two vs. Two), using DST is not an appropriate approach to increase the overall accuracy or κ value. This is because, TVT produces completely different results than OVR or OVO. For TVT, the test result of a sample predicts it to belong into one of the combination of two classes. This means that, it does not predict one single class for the test sample. On the other hand, OVO approach predicts whether the test sample belongs to one of the two classes it is looking into and for OVR approach, one class is checked against rest of the classes. Therefore, it can be clearly seen that all three approaches are different from each other and they cannot be combined using DST, as their predicted results are different from each other.

Table I shows that the research work that have been carried out so far, reports the maximum average κ value to be 0.70 [1] for OVR approach. Only in [6], the authors have used other approaches than OVR, i.e. PW and DC. In this paper, we have addressed both the OVR and PW approaches and compared our proposed method's performances against the state-of-the-art methods. This paper also shows that with smaller number of selected channel combinations and with a specific frequency sub-band, better performances can be achieved while using OVR and PW/OVO approaches.

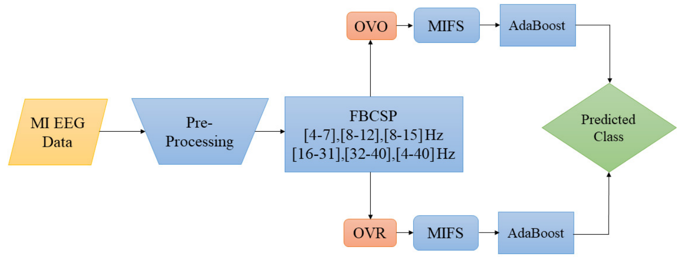

Fig. 1. Flow diagram of the proposed method

III. Proposed Methodology

This paper proposes a MI-task separation technique using FBCSP for feature extraction and AdaBoost for classification. A set of pre-processing steps are introduced over here in order to improve the signal-to-noise ratio (SNR). A schematic flow diagram for the proposed scheme is depicted in Fig.1. Following are the details of the EEG dataset and all the methods that have been used for this work.

A. MI Dataset and Protocol

All the experiments for performance evaluation are performed on the BCI competition IV dataset 2a. This dataset contains 4-class MI EEG data from 9 subjects, namely, left hand, right hand, feet and tongue's imaginary movements. There are two sessions of data, one each for training and testing/evaluation. Each of the session contains 288 trials of data (22 channels) recorded from each subject. For experimental purpose, we have considered the data starting from 0.5s of visual cue to 3.5s of MI task. Therefore, a total of 3s window is used for both training and testing purposes.

B. Pre-processing

At the beginning of both training and testing phases, a set of pre-processing steps have been carried out on the MI EEG data [17] in order to increase the signal to noise ratio. First of all, a common average referencing (CAR) filter is used on all the 22 channels to minimize the artifacts related to the choice of unsuitable references. These CAR-filtered signals are then down-sampled by a factor of two from 250Hz and subsequently z-score normalized. Finally, detrending method is applied on all the 22 channels by individually removing their best-fit line. All these steps allow us to concentrate only on the data fluctuation about their estimated trend.

C. FBCSP for Feature Extraction

Training and testing features are extracted from 12 EEG channels, (FC_1,FC_z,FC_2,C_3,C_1,C_z,C_2,C_4,CP_1,CP_z,CP_2,P_z). This particular channel combination have been chosen by keeping the central region of the brain in focus and by consulting various state-of-the-art research articles [11], [14]. FBCSP comprises multiple band-pass filters which uses zero-phase Chebyshev Type II filters and spatial filters that are used for CSP feature extraction. Initially, a filter bank with several sub-band filters are employed to find the EEG signals' measurements into these specific sub-bands. After that, each of these sub-bands are subjected to CSP feature extraction. Then, typically a feature selection method, i.e. mutual information-based feature selection (MIFS), is employed in order to select the most discriminative feature which are generated by different filter banks. For this particular experiment, we have used six different frequency sub-bands, namely, $[4-7]$Hz, $[8-12]$Hz, $[8-15]$Hz, $[16-31]$Hz, $[32-40]$Hz and $[4-40]$Hz.

D. Training and Testing Strategies

Training and testings are performed using two different approaches: PW/OVO and OVR. In PW/OVO approach, FBCSP features are extracted for separating every pair of classes, i.e. left hand and right hand, left hand and feet, etc. For the 4-class MI, 6 binary classifiers are required to discriminate each individual classes. Majority voting scheme is used for predicting the final class from the predicted classes of all the binary classifiers. Similarly for OVR approach, FBCSP features are extracted from each of the classes while discriminating them from rest of the classes. Therefore, a total of 4 classifiers are required for this approach.

E. AdaBoost for Classification

Adaptive Boosting or AdaBoost converts a set of weak classifiers into a stronger one. The classification using AdaBoost can be represented as Eq.(1):

$$C(x) = \pm \left(\sum_{n=1}^{K} w_n \, c_n \, (x) \right), \tag{1}$$

where c_n is the n-th weak classifier and w_n is its corresponding weight and K is the total number of classes. $\pm$ in Eq.(1) denotes positive or negative class and $C(x)$ stands for final chosen classifier of x. These weak classifiers are actually decision trees and the AdaBoost puts more weight on difficult to classify instances and less weights on those that are already classified well. For our experiments, we have used the extracted FBCSP features, as described in section III-C, to train the AdaBoost classifier. Matlab's built-in function (*fitensemble*) have been used, with 0.1 as learning rate for all the training and 10^5 as the number of epochs.

TABLE II
COMPARISON OF κ VALUES FOR THE OVA APPROACH BETWEEN PROPOSED AND STATE-OF-THE-ART METHODS.

Paper	Subjects									Average
	S1	**S2**	**S3**	**S4**	**S5**	**S6**	**S7**	**S8**	**S9**	
[7]	0.66	0.42	0.77	0.51	0.50	0.21	0.30	0.69	0.46	0.50
[10]	0.76	0.32	0.76	0.47	0.31	0.34	0.59	0.76	0.74	0.56
[6]	0.68	0.42	0.75	0.48	0.40	0.27	0.77	0.75	0.61	0.57
[9]	0.74	0.38	0.72	0.50	0.26	0.34	0.69	0.71	0.76	0.57
[8]	0.67	0.49	0.77	0.59	0.52	0.31	0.48	0.75	0.65	0.58
[18]	0.56	0.41	0.43	0.41	0.68	0.48	0.80	0.72	0.78	0.59
[11]	0.75	0.50	0.74	0.40	0.19	0.41	0.78	0.72	0.78	0.59
[12]	0.74	0.35	0.76	0.53	0.38	0.31	0.84	0.74	0.74	0.60
[13]	0.70	0.33	0.74	0.39	0.54	0.32	0.75	0.79	0.76	0.59
[19]	0.59	0.41	0.82	0.57	0.38	0.29	0.79	0.80	0.72	0.60
[20]	0.76	0.38	0.87	0.60	0.46	0.34	0.77	0.76	0.74	0.63
[14]	0.69	0.51	0.87	**0.85**	**0.78**	0.42	0.54	**0.97**	0.45	0.66
[1]	0.83	0.51	**0.88**	0.68	0.56	0.35	**0.90**	0.84	0.75	0.70
[15]	0.63	0.43	0.74	0.54	0.19	0.26	0.63	0.62	0.69	0.53
[16]	0.86	0.24	0.70	0.68	0.36	0.34	0.66	0.75	**0.82**	0.60
Proposed	**0.95**	**0.71**	0.78	0.67	0.63	**0.77**	0.69	0.66	0.74	**0.73**

F. Performance Metrics

To test the performance of our proposed scheme, the κ values (Eq.(2)) [1] for individual subjects and their mean are calculated.

$$\kappa = \frac{Pc_a - Pc_e}{1 - Pc_e} \tag{2}$$

$$Pc_e = \frac{1}{N_t^2} \sum_{k=1}^{K} n_k^1 \, n_k^2, \tag{3}$$

where, Pc_a represents the accuracy or observed overall agreement on test samples and Pc_e stands for the chance agreement. N_t in Eq.(3) is the total number of testing trials, K is the total number of classes, n_k^1 is the number of times the proposed algorithm predicted class k and n_k^2 is the total number of actual class k in the testing trials.

IV. RESULTS & DISCUSSION

This section presents the experimental results which have been achieved on BCI competition IV dataset 2a for MI task separation and finds the best suitable frequency sub-band for the same purpose. First, the most significant sub-band is determined using both OVR and PW/OVO approaches. For this, we have calculated the κ-values for six sub-bands, namely, $[4-7]$Hz, $[8-12]$Hz, $[8-15]$Hz, $[16-31]$Hz, $[32-40]$Hz and $[4-40]$Hz. It is found that for PW/OVO approach, both $[8-15]$Hz and $[4-40]$Hz sub-bands perform better than the others, while $[4-40]$Hz being slightly better than the $[8-15]$Hz. Similarly for OVR approach, the results of $[8-15]$Hz sub-band is better than the $[4-40]$Hz. Therefore, it can be concluded that for MI task, the α sub-band, i.e. $[8-15]$Hz is the most dominant one. These results can be verified through Fig.2. Although, $[4-40]$Hz sub-band's performance is also at par with $[8-15]$Hz, therefore with only $[8-15]$Hz sub-band, MI-tasks can be separated with good accuracy.

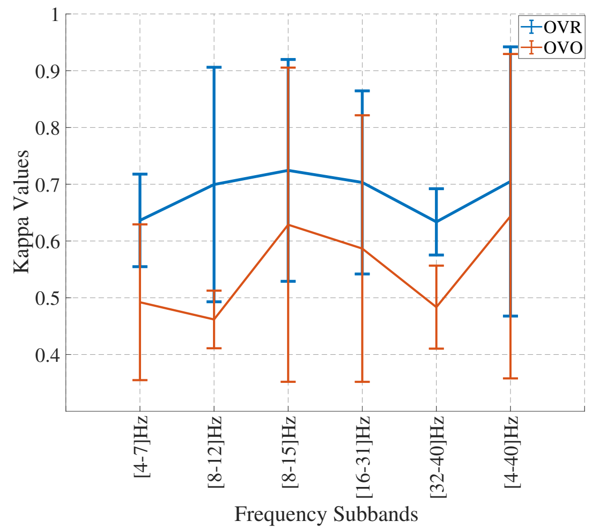

Fig. 2. κ values for different sub-bands on BCI competition IV dataset 2a for OVO and OVA approaches.

TABLE III
COMPARISON OF κ VALUES FOR THE PW/OVO APPROACH BETWEEN PROPOSED AND STATE-OF-THE-ART METHODS.

Paper	Subjects									Average
	S1	**S2**	**S3**	**S4**	**S5**	**S6**	**S7**	**S8**	**S9**	
[6]	0.782	0.407	0.755	0.528	0.417	0.185	0.796	0.741	0.537	0.572
Proposed	**0.958**	**0.636**	0.718	0.472	**0.510**	**0.692**	0.602	0.561	**0.663**	**0.646**

Tables II and III show the results for OVR and OVO approaches, respectively. For OVR approach, our proposed method is able to achieve the best κ values for subjects 1, 2 and 6 in comparison with all the existing state-of-the-art results. An average κ value of 0.73 has been achieved which is the highest to-date, to the best of our knowledge. For the OVO approach, as per the literature, there is no other article than Ang et al. [6], where the PW performances are reported. Therefore, we only compared our method with [6] and found that for subjects' 1, 2 and 6, significant improvement in the κ values are achieved (Table III). Similarly, the average κ value (0.646) is also much higher (0.646 against 0.572) than that in [6].

Many researchers have shown in the past that decision tree or random forest technique performs better than the other machine learning techniques for MI based EEG signal classification [21], [22]. The motivation behind using an AdaBoost classifier came from the fact that it creates a strong classifier by combining many weak classifier such as decision trees. This is achieved by initially creating a model from the training data and then by creating another model to try to correct the errors that have been produced by the first model. Moreover, AdaBoost has a high degree of precision and does not require much tweaking for its hyper-parameters. Theoretically, it is less prone for overfitting as its parameters are not jointly optimized. The obtained results has justified these facts and the combination of FBCSP for feature extraction and AdaBoost for classification have been able to outperform the existing state-of-the-art techniques.

V. Conclusions

This paper proposes a MI-task separation method based on FBCSP features and AdaBoost classification technique. Initially, a set of pre-processing steps have been used in the MI-task elicited EEG signal of BCI IV 2a dataset. The proposed method displays a significant improvement in performance over the existing state-of-the-art techniques, in terms of the κ values. This paper also shows the performance comparison of different frequency sub-bands for separating different MI-tasks. Results show that the α sub-band ($[8-15]$Hz) is the best suited one for the MI-task separation.

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